Computer Graphics - Rendering

Course Structure - Ray Tracer Implementation

- [ ] Module 1 - **due** Oct 17th
- [ ] Module 2 - **due** Oct 31st
- [ ] Module 3 - **due** Nov 21st
- [ ] Module 4 - **due** Dec 5th

Course website: https://ksubr.github.io/CGR2026/#Coursework

Textbook

- https://www.pbr-book.org/
- https://raytracing.github.io/books/RayTracingInOneWeekend.html

---

Computer Graphics Rendering — Complete Lecture Notes

Part I: Cameras and Optics

CGR25_L3

1. Pinhole Camera Model

Light travels in straight lines through a single point (pinhole) onto an image plane, producing an inverted image. The **virtual pinhole model** places the image plane in front so the image is upright.

Projection pipeline:

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} f_x & \alpha & u \\ 0 & f_y & v \\ 0 & 0 & 1 \end{bmatrix}}_{\text{Intrinsic } K} \cdot \underbrace{\begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{bmatrix}}_{\text{Extrinsic } [R|t]} \cdot \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

- **Intrinsic** K: focal length in pixels (f_x, f_y), skew α, principal point (u, v).
- **Extrinsic** $[R|t]$: rotation R and translation t of camera in world space.

> 中文注释：内参矩阵把相机坐标映射到像素坐标，外参矩阵把世界坐标映射到相机坐标。两者相乘就是完整的投影矩阵。

2. Lenses, Aperture, and Depth of Field

Pinhole trade-off: small hole → sharp but dark; large hole → bright but blurry. A **lens** solves this by bending rays so they converge.

Thin lens formula:

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

- f = focal length, u = object distance, v = image distance.
- **Depth of Field (DoF)**: range of depths that appear sharp.
- **Aperture (f-number)**: $N = f/D$. Small N → large aperture → shallow DoF, more light.

> 中文注释：光圈越大（f 值越小），景深越浅（背景越模糊），进光量越多。

3. Digital Sensors

- Sensor = grid of "buckets" (photosites) converting photons to electrons.
- **CCD**: lower noise, sequential readout, blooming/smear artifacts.
- **CMOS**: faster, lower power, rolling shutter distortion.
- Common artifacts: read-out noise, rolling shutter, motion blur.

Part II: 3D Modelling and Materials

CGR25_L5

1. Modelling Approaches

Approach	Description	Examples
Manual	Human artists sculpt/model	Maya, Blender, ZBrush
Mathematical	Procedural / simulation	Physics sim, L-systems
Physical measurement	3D scanning, motion capture	Photogrammetry

2. Shape Representations

- **Explicit**: meshes (vertices/edges/faces), point clouds, voxels.
- **Implicit**: $f(x, y, z) = 0$ — SDFs, parametric surfaces.
- **CSG (Constructive Solid Geometry)**: combine primitives via union, intersection, difference.

> 中文注释：显式表示直接给出表面（如三角形网格），隐式表示用数学方程定义形状。CSG 用布尔运算组合简单几何体。

3. Production Pipeline

Modelling → Texturing → Rigging → Animation → Lighting → Rendering.

4. Materials

- Defined by **measured data**, **parametric models** (GGX, Cook-Torrance), or **procedural generation**.
- Key properties: albedo, roughness, metallic, normal maps, displacement maps.

Part III: Raytracing and Rendering

CGR25_L7, L8

1. Ray Casting

For each pixel:

1. Cast a ray from camera through the pixel.

2. Find closest intersection with scene.
3. Shade the hit point.

Ray-sphere intersection: solve $|O + tD - C|^2 = r^2$ for t (quadratic).

2. Acceleration Structures

- **BVH** (Bounding Volume Hierarchy): tree of bounding boxes — most common.
- **Grids**, **octrees**, **bounding boxes**.
- GPUs process **>10 billion rays/second** in parallel.

> 中文注释：加速结构避免每条光线测试所有物体。BVH 用包围盒树逐层剔除，现代 GPU 可并行处理百亿级光线。

3. Shading — Blinn-Phong

$$I_p = k_a i_a + \sum_m \left(k_d (\hat{L}_m \cdot \hat{N}) i_{m,d} + k_s (\hat{H}_m \cdot \hat{N})^\alpha i_{m,s} \right)$$

- **Ambient** + **Diffuse** (Lambertian) + **Specular** (highlight).
- $\hat{H} = (\hat{L} + \hat{V}) / |\hat{L} + \hat{V}|$ (halfway vector, Blinn-Phong).

4. Advanced Light Transport

Secondary rays simulate:

- **Shadows** (shadow ray to light)
- **Reflection** (mirror bounce)
- **Refraction** (transparent materials)
- **Global illumination** (multiple bounces)

Part IV: Transformations

CGR25_L9

1. Classes of Transformations

Class	Properties	Examples
Linear	Preserves origin	Scale, rotation, shear
Affine	Linear + translation	All above + translation
Projective	Preserves lines	Perspective
Rigid/Euclidean	Preserves distances	Translation, rotation
Similitudes	Rigid + uniform scale	+ uniform scaling

2. Homogeneous Coordinates

Represent 3D points as $(x, y, z, 1)$ so translation becomes matrix multiplication. Composition is **not commutative** — order matters.

3. Correct Normal Transformation

Incorrect: $n_{WS} = M n_{OS}$ (only works for similitudes).
Correct: $n_{WS} = (M^{-1})^T n_{OS}$.

Derivation: tangent v satisfies $n^T v = 0$. After transform $v_{WS} = M v_{OS}$, require $n_{WS}^T v_{WS} = 0$:

$$(n_{OS}^T M^{-1})(M v_{OS}) = 0 \implies n_{WS}^T = n_{OS}^T M^{-1} \implies n_{WS} = (M^{-1})^T n_{OS}$$

> 中文注释：法线不能直接用模型矩阵变换（非均匀缩放会扭曲）。正确做法是用模型矩阵的逆转置。

Part V: Monte Carlo Integration

CGR25_L12

1. Why Randomness?

High-dimensional integrals (over pixels, lens, time, materials) are intractable with deterministic quadrature. Monte Carlo error is $O(1/\sqrt{N})$ regardless of dimension.

Estimator:

$$\int f(x)\,dx \approx \frac{1}{N}\sum_{i=1}^{N}\frac{f(x_i)}{p(x_i)}, \quad x_i \sim p(x)$$

- **Unbiased**: expected value = true integral.
- **Variance** decreases with N; **importance sampling** reduces it further.

> 中文注释：蒙特卡洛用随机采样估计积分，误差按 $1/\sqrt{N}$ 下降（与维度无关）。重要性采样让 $p(x)$ 接近 $f(x)$ 以降低方差。

2. Buffon's Needle & Monte Carlo Origins

- Buffon (1777): estimate π by dropping needles on lined floor.
- Ulam & von Neumann (1940s): Monte Carlo method for nuclear physics.

Part VI: Path Tracing

CGR25_L13

1. From Whitted Raytracing to Path Tracing

Whitted: only specular reflection/refraction, hard shadows, no diffuse inter-reflection.
Path tracing (Kajiya 1986): solves the full rendering equation via Monte Carlo.

2. Light Transport Fundamentals

Radiance $L(x,\omega)$: $W\cdot m^{-2}\cdot sr^{-1}$ — power per projected area per solid angle.
Irradiance $E(x)$: $W\cdot m^{-2}$ — power per area.

BRDF $f_r(x,\omega_i,\omega_o)$: ratio of outgoing radiance to incoming irradiance.

- **Reciprocity**: $f_r(\omega_i,\omega_o)=f_r(\omega_o,\omega_i)$.
- **Energy conservation**: $\int f_r\cos\theta\,d\omega \le 1$.

Measurement: gonioreflectometer → 4D tabulated data.

Common models: Lambertian, Phong/Blinn-Phong, Cook-Torrance, GGX, Disney BRDF.

> 中文注释：BRDF 描述"从某方向入射的光有多少被反射到另一方向"，满足互易性和能量守恒。测量需要专门的测角反射计。

3. The Rendering Equation (Kajiya 1986)

$$\boxed{L_o(x, \omega_o) = L_e(x, \omega_o) + \int_\Omega f_r(x, \omega_i, \omega_o)\, L_i(x, \omega_i)\, \cos\theta_i \, d\omega_i}$$

- **Recursive**: L_i at one point = L_o from another.
- **Citations**: >60,000 (Google Scholar).

> 中文注释：渲染方程是图形学的核心。出射光 = 自发光 + 所有入射光被 BRDF 反射的积分。它是递归的，因为入射光本身来自其他点的出射光。

4. Solving via Path Tracing

Algorithm:

1. Sample hemisphere at last bounce to camera.
2. Trace sample ray to intersection.
3. Sample those hemispheres recursively until k bounces.
4. Estimate radiance by averaging.

Cost: n^k samples per pixel (e.g., 20 samples, 3 bounces = 8000 spp).

Random sampling at each level: instead of branching, randomly pick one direction per bounce — trace many independent paths and average.

> 中文注释：路径追踪从相机发射大量随机路径，每条路径随机弹射，最后取平均。避免了指数爆炸。

5. Distributed Ray Tracing (Cook et al. 1984)

Random sampling simulates:

Effect	Sampled dimension
Depth of field	Lens aperture (2D)
Motion blur	Time (1D)

Effect	Sampled dimension
Soft shadows	Area light (2D)
Glossy reflection	BRDF lobe (2D)

Ray tree: 3 bounces, 3 samples → $3^3 = 27$ rays; 10 samples → $10^3 = 1000$.

6. Terminating Paths — Russian Roulette

- Choose termination probability q.
- With probability q, terminate (return 0).
- With probability $1 - q$, continue and **divide throughput by** $1 - q$.
- **Unbiased**.

> 中文注释：俄罗斯轮盘赌随机终止低贡献路径，但把存活路径除以存活概率，保持无偏。

7. Path Tracing — Maths

$$I_p = \int_{\mathcal{P}} h_p(X) \cdot f(X)\, dX$$

Path contribution:

$$f(X) = L_e(x_1) T(x_1, x_2) G(x_1, x_2) \cdot \left(\prod_{i=1}^{k-1} f_r(x_i) T(x_i, x_{i+1}) G(x_i, x_{i+1}) \right) \cdot W(x_k)$$

- T: visibility; G: geometry term $\cos\theta_i \cos\theta_o / \|x_i - x_{i+1}\|^2$; W: camera response.

8. Importance Sampling Paths

- **Sample light source**: good for small bright lights.
- **Sample BRDF**: good for glossy surfaces.
- **MIS (Multiple Importance Sampling)**: combine both, weighted by probability.

Balance heuristic: $w_i(x) = p_i(x) / \sum_j p_j(x)$.

> 中文注释：MIS 结合光源采样和 BRDF 采样，按生成概率加权，取长补短。

Part VII: Sampling and Reconstruction

CGR25_L14

1. Sampling & Reconstruction Problem

Convert continuous $f(x)$ to discrete samples; reconstruct via **convolution with a kernel** $g(x)$.

2. Dirac-Delta Distribution

Properties: zero except at 0; unit integral; identity for convolution.

Sampling function: $S(x) = \sum_k \delta(x - x_k)$.

3. Fourier Analysis

Eigenfunction of derivative: $\frac{d}{dx} e^{\lambda x} = \lambda e^{\lambda x}$.
Fourier transform:

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i \omega x} \, dx$$

Interpretation: projection onto cos and sin; frequency-domain representation.

> 中文注释：傅里叶变换把函数分解成不同频率的正弦波。$e^{\lambda x}$ 是微分算子的特征函数。

4. Fourier Duals (Key Pairs)

Spatial	Frequency
$\text{rect}(ax)$	$\frac{1}{\|a\|}\text{sinc}(\omega/a)$
$\text{sinc}(ax)$	$\frac{1}{\|a\|}\text{rect}(\omega/a)$
$e^{-a\|x\|}$	$\frac{2a}{a^2+4\pi^2\omega^2}$

Narrow in space → wide in frequency (and vice versa).

5. Convolution Theorem

$$f * g \leftrightarrow \hat{f} \cdot \hat{g}$$

Convolution in spatial domain = multiplication in frequency domain. Enables FFT-based fast convolution.

6. Sampling in Fourier Domain

Sampling = multiplication by $S(x)$ → convolution with $\hat{S}(\omega)$ in frequency domain → **replication** of spectrum at intervals $1/T$.

Aliasing: if replicas overlap (function not band-limited), high frequencies masquerade as low frequencies.

> 中文注释：采样在频域表现为频谱的周期性复制。如果原函数不是带限的，复制会重叠，产生混叠（aliasing）。

7. Nyquist-Shannon Sampling Theorem

> If $x(t)$ has no frequencies above B Hz, it is completely determined by samples spaced $1/(2B)$ seconds apart.

Nyquist rate: $2B$. To avoid aliasing: sample at ≥ $2B$, or low-pass filter first.

8. Reconstruction

Ideal: multiply spectrum by box → convolve with **sinc** in spatial domain.
Practical kernels: box (nearest), tent (linear), Gaussian, Mitchell-Netravali.

Anti-aliasing: supersampling, pre-filtering, stochastic sampling.

> 中文注释：理想重建核是 sinc（频域矩形），但它是无限长的，实际用 box、tent、Gaussian 等。抗锯齿可以超采样或预滤波。

Part VIII: Fast Rendering Pipelines (Shaders)

CGR25_L17

1. Evolution of Graphics Pipelines

1992 — Fixed-function: Vertex Transform & Lighting → Primitive Setup & Rasterization → Fragment Coloring & Texturing → Blending.

2010 — Programmable: Vertex Shader → Tessellation → Geometry Shader → Rasterizer → Fragment Shader → Blending.

Modern pipelines:

- **Raster**: Input Assembler → Vertex → Hull → Tessellation → Domain → Geometry → Rasterizer → Pixel → Output Merger.
- **Compute**: Input → Compute Shader → Output (GPGPU).
- **Ray Tracing**: Raygen → Scene Traversal → Intersection → Any Hit → Closest Hit → Miss → Output.
- **Mesh Geometry**: Amplification → Mesh → Input Assembler → Rasterizer → Pixel → Output Merger.

> 中文注释：图形管线从固定功能发展到可编程着色器。现代管线支持光追、计算、网格着色器等专用阶段。

2. Shader Examples

Vertex Shader (GLSL) — transform position, compute lighting:

```
gl_Position = PMatrix * ecPosition;
ecNormal = normalize(vec3(modelView * vec4(MCNormal, 0.0)));
```

Geometry Shader (GLSL) — emit primitives:

```
layout(triangles) in;
layout(triangle_strip, max_vertices = 3) out;
```

Fragment Shader (GLSL) — per-pixel color:

```
gl_FragColor.x = (1.0 - weight) * x + weight * (-f - diag_14 * x_14 + ...);
```

Metal (Apple):

```
vertex VertexOut vertexMain(VertexIn in [[stage_in]]) { ... }
fragment half4 fragmentMain(VertexOut in [[stage_in]], texture2d<float> tex
[[texture(0)]]) { ... }
```

3. Shader Effects

- **Vertex shader with time**: deform geometry (sphere → heart shape).
- **Fragment shader**: procedural gradients, dissolve, water ripples.
- **ShaderToy**: online GLSL playground.

> 中文注释：顶点着色器可以随时间变形几何体，片段着色器可以生成程序化效果。ShaderToy 是学习着色器的好平台。

Part IX: Machine Learning in Computer Graphics

CGR25_L18

1. Classical ML

- **Supervised**: classification (category), regression (number).
- **Unsupervised**: clustering (similarity), association (dependencies), dimensionality reduction.
- **Curve-fitting methods** (XKCD): linear, quadratic, loess, logistic, piecewise, ad-hoc, house-of-cards.

2. Interpolation (Regression)

- 1D: nearest, linear, cubic.
- 2D: nearest, bilinear, bicubic.
- **Overfitting**: high-degree polynomial oscillates.
- **Fitting with basis**: $y(x) = \sum_i w_i \phi_i(x)$ — determine weights.

> 中文注释：过拟合是模型太复杂，完美拟合训练数据但泛化差。基函数方法把函数表示为基函数的加权和。

3. Neural Networks

Single layer:

$$y = \sigma(W_2 \cdot \sigma(W_1 x + b_1) + b_2)$$

Sigmoid: $\sigma(x) = \frac{1}{1-e^{-(wx+b)}}$.

Architectures: FC, CNN, GAN, Transformer, Diffusion, Foundation models.

4. NeRF (Neural Radiance Fields)

Paper: Mildenhall et al., 2021 (60,000+ citations).
Idea: Represent scene as $F_\Theta : (x, y, z, \theta, \phi) \to (RGB, \sigma)$.

- **Training**: render images, compare to ground truth, backprop.
- **Rendering**: ray march, accumulate color and density.

> 中文注释：NeRF 用神经网络表示 3D 场景（输入位置+方向，输出颜色+密度）。训练时从不同视角渲染并与真实图像比较。渲染用体积渲染方程。

5. 3D Gaussian Splatting

Paper: Kerbl et al., SIGGRAPH 2023.
Idea: Scene = set of 3D Gaussians (position, covariance, opacity, SH color). Project to 2D, sort, alpha-blend. Real-time (100+ FPS).

> 中文注释：3D Gaussian Splatting 用一组 3D 高斯球表示场景，渲染时投影混合。比 NeRF 快很多。

6. Dimensionality Reduction

- **PCA** (linear), **t-SNE/UMAP** (non-linear), **Autoencoders**.
- **Manifold of fonts (GPLVM)**: interpolate between fonts.

7. Generative Models

- **VQGAN**: text-to-image.
- **Diffusion models**: forward (add noise) + reverse (denoise).
- **DiLightNet** (SIGGRAPH 2024): lighting control for diffusion.
- **DiffusionRenderer** (CVPR 2025): delighting/relighting.

8. Neural Rendering

- **N-BVH** (SIGGRAPH 2024): neural ray queries with BVH, multi-resolution hash grid + MLP.
- **RenderFormer** (SIGGRAPH 2024): transformer-based rendering with global illumination.

9. Denoising Path-Traced Images

Traditional pipeline: materials/geometry → sampled rays → hand-tuned denoisers → composite (low-res) → upscale.
Limitations: inaccurate lighting, accumulated bad pixels, lost high-frequency data.

DLSS 3.5 (AI Ray Reconstruction):

- Input: materials, high-frequency rays, motion vectors.
- Process: ray reconstruction + super resolution → temporal feedback → optical flow → frame generation.
- Output: 4K high-quality RT image.
- Trained on 5× more data than DLSS 3.

Performance: DLSS off = 21 FPS; DLSS 2 = 62 FPS; DLSS 3 = 102 FPS; DLSS 3.5 = 113 FPS.

> 中文注释：DLSS 3.5 用 AI 重建光线追踪图像，从低分辨率噪声图像恢复高质量 4K 图像，速度提升 5 倍以上。

Part X: VR and Conclusion

CGR25_L19

2. Uncanny Valley (Mori 1970)

Human likeness vs. likability: rises with cartoon/robot, dips sharply for near-human (uncanny valley), then rises to human.

> 中文注释：恐怖谷理论：当机器人/虚拟人越来越像人但又不完全像时，会让人感到诡异和不适。

3. VR Systems

Component	Description
Stereo vision	Two eyes → parallax → depth
Display	Micro-display + magnifying lens
Haptics	Gloves, suits, force feedback
Audio	Stereo/surround sound
Smell	VR smell devices

4. Depth Perception

- **Vergence**: eye rotation to converge on object.
- **Accommodation**: lens focusing.
- **Conflict in flat stereo displays**: vergence and accommodation disagree.

> 中文注释：真实世界中，眼睛的会聚（vergence）和调焦（accommodation）是一致的。平面立体显示器中两者冲突，导致视觉疲劳。

5. Stereo Displays

Method	Description
Side-by-side	Wall-eyed / cross-eyed
Stereoscopes	19th century
Anaglyphs	Red/cyan glasses
Polarisation	Linear/circular
Active shutters	Eclipse glasses
Interference	Chromatic
Autostereoscopy	Lenticular, parallax barrier

6. HMDs (Head-Mounted Displays)

- **Most HMDs are simple magnifiers**: micro-display + lens.
- **Stereo rendering**: off-axis projection (correct) vs. toe-in (incorrect, creates vertical parallax).
- **Lens distortion**: pin-cushion (pincushion) and barrel distortion.

> 中文注释：HMD 本质是放大镜 + 微显示器。立体渲染必须用离轴投影（off-axis），不能用 toe-in（会导致垂直视差）。镜头畸变需要预补偿。

7. Apple Vision Pro (2024)

- 23 million pixels, 3D display system.
- Micro-OLED, 7.5-micron pixel pitch.
- 92% DCI-P3, 90/96/100 Hz refresh rates.
- Stereoscopic 3D camera, LiDAR, 6 world-facing cameras, 4 eye-tracking cameras.

Summary Table — Key Concepts

Topic	Key Idea	Key Equation
Pinhole Camera	3D → 2D projection	$K[R\|t]$
Thin Lens	Focal relationship	$1/f = 1/u + 1/v$
Depth of Field	Aperture controls blur	$N = f/D$
Modelling	Manual, procedural, scanned	Meshes, voxels, SDFs
Ray Casting	Ray-scene intersection	$t = \frac{N \cdot (P_0 - O)}{N \cdot D}$
Blinn-Phong	Ambient + diffuse + specular	$k_a i_a + k_d(\hat{L} \cdot \hat{N}) i_d + k_s(\hat{H} \cdot \hat{N})^\alpha i_s$
Rendering Equation	Recursive light transport	$L_o = L_e + \int f_r L_i \cos\theta \, d\omega$
Monte Carlo	Random sampling estimator	$\frac{1}{N} \sum f(x_i)/p(x_i)$

Topic	Key Idea	Key Equation
Path Tracing	Trace random paths, average	$I_p = \int h_p(X) f(X)\, dX$
Russian Roulette	Unbiased path termination	Divide by $1 - q$
Nyquist-Shannon	Sampling rate ≥ 2B	$1/(2B)$ spacing
Shaders	Programmable GPU stages	Vertex, geometry, fragment
NeRF	Neural radiance field	$F_\Theta : (x, y, z, \theta, \phi) \to (RGB, \sigma)$
3D Gaussian Splatting	Real-time radiance fields	3D Gaussians + alpha blending
DLSS 3.5	AI ray reconstruction	21 → 113 FPS
VR	Stereo + head tracking + haptics	Vergence-accommodation conflict